Class 8:

1. Find the volume common to the cylinders $x^2 + y^2 = a^2$ and $x^2 + z^2 = a^2$ ans: $\frac{16a^3}{3}$
2. Find the volume cut from the sphere $x^2 + y^2 + z^2 = a^2$ by the cone $x^2 + y^2 = z^2$ above XY plane.

   ans: $\frac{\pi a^3}{3}\left(2-\sqrt{2}\right)$
3. Find the average value of $f(x, y, z) = x + y + z$, using triple integrals over the region D= $\{(x, y, z) \mid 0 \le x \le 1, 0 \le y \le 3, 0 \le z \le 5\}$. ans: $\frac{9}{2}$

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